

Unit 4: Key Types of Business Data Analytics: Descriptive, Predictive and Prescriptive

Assignment 8: Predictive Analytics: A Comprehensive Study for Business Data Analytics using Python

Topics Covered: Classification (Logistic Regression/ kNN / Decision Tree/ Random Forest/ SVM); Splitting Dataset (Training, Testing); Model Evaluation (Accuracy, Precision, Mean Squared Error, Mean Absolute Error, Recall, F1-Score, Confusion Matrix, Classification Report)

The assignment aims to provide a comprehensive guide to predict future outcomes in a business context using various classifiers.

Data Preparation:

Collect historical data of customers, including demographic information, purchase history, product preferences, churn rate. Preprocess the data by handling missing values, encoding categorical variables, and splitting the dataset into training and testing sets.

Classification Model Training:

Implement a classifier model using Python's scikit-learn library. Train the classifier on the training data to predict customer preferences, future purchase behaviour or churn rate.

Model Evaluation:

Evaluate the performance of the classifier using appropriate evaluation metrics such as accuracy, precision, recall, F1-score, mean_squared_error, confusion matrix, classification report.

Students are required to create a folder as FourDigitsRollno_Roomno_DeptCode_BDA (For Example, 0012_14_COMA_BDA) on their desktop. All the files related to the assignment should be saved within this folder. Students are required to use either Spyder or Jupyter Notebook to complete the assignment. The final working code should be saved as AssignmentNo_Rollno.py

Customer Churn Prediction with Classifiers (Predictive Analytics)

1. A telecom service provider aims to predict customer churn and identify the most effective machine learning model for proactive retention planning. The company uses historical customer data that includes service usage behavior and subscription details collected over a one-year period. In the dataset cust_churn.csv, each record represents an individual customer along with a churn indicator.

Data Preparation

- a. Load the dataset using the read_csv() function from the Pandas library.
- b. Identify and separate the independent variables (features) and the dependent variable (Churn).
- c. Display the selected features and target variable.

Train-Test Split

- d. Split the dataset into training and testing sets, using 80% of the data for training and 20% for testing.

Model Implementation

- e. Build and train the following machine learning classifiers to predict customer churn:
 - Logistic Regression (*set maximum iterations = 1000*)
 - k-Nearest Neighbors (kNN)
 - Decision Tree Classifier
 - Random Forest Classifier
 - Support Vector Machine (SVM)

Model Evaluation

- f. For each classifier, compute and display:

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- Accuracy
- Precision
- Recall
- F1-Score
- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Confusion Matrix
- Classification Report
